CASE STUDY QUESTION

Q4 [Exploration] Relationship Investigatio

- Using **Cognos Exploration**, explore the relationship between Shift Type (Morning / Afternoon / Night) and Defect Rate. Investigate whether specific supervisors or assembly lines are associated with higher defect rates.
- Also explore the correlation between Raw Material Supplier and Scrap Units. Present key findings with visualizations.

CASE STUDY
Analyzing Defect Rate Correlations

Using Cognos Exploration, analyze defects to identify key correlations based on shift schedules, supervisors, and raw material suppliers. Present key insights with visualizations, ensuring findings are clear and actionable.

Defect Rate Correlation Analysis

Key Insights with Visualizations

01 Explore Shift Type vs. Defect Rate

Analyze the defect rate by shift type (Morning, Afternoon, Night) within each assembly line.

02 Identify High-Risk Supervisors/Lines

Identify the supervisors and assembly lines with the highest defect rates for targeted improvement.

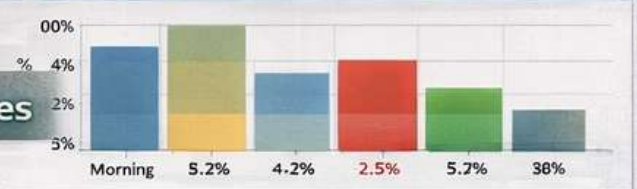
03 Correlate Raw Material Supplier & Scrap Units

Examine the relationship between Raw Material Supplier and Scrap Units/ Defect Rate. Discover which suppliers are associated with higher defect rates and scrap production.

04 Present Key Findings

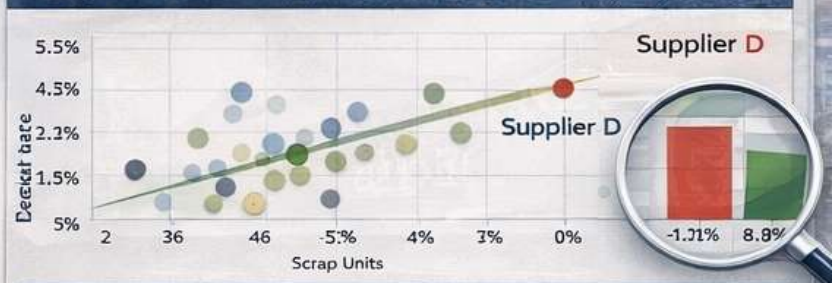
- Summarize key findings with clear visualizations and bullet points for actionable insights.

Defect Rate by Shift Type



Supervisor	Morning	Defenor Units	Night	Brown Sare
Johnson	290	4.90	6.4%	6.8%
Williams	155	3.50	6.8%	1.9%
Davis	240	230	8.5%	1.9%
Brown	255	190	9.2%	1.9%
Miller	140	220	1.2%	1.8%

Scrap Units & Defect Rate by Raw Material Supplier



Defect Rate by Raw Material Supplier		
Supplier A	8.2%	0.9%
Supplier B	6.2%	8.8%
Supplier E	5.2%	4.1%

04 Key Findings

- Night shift on Assembly Line 2 has the highest defect rate (6.4%).
- Supervisor Davis on Assembly Line 2 has the highest defect rate, 7.1%
- Supplier D shows a strong correlation between high scrap units and high defect rate.

My IBM | Develop concep | IBM Cognos An | My IBM | My CS 4 | My CS 3 | Ask Gemini

ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=i72AE601D97324CF9A8A954B0B7685710&objRef=i72AE601D97324CF9A8A954B0B768571...

IBM Cognos Analytics | My CS 4

Create | 1/9 | Table | Related | Compare | Analytics | Details | Fields | Properties

Cards

1 Raw Material Su...nits Defective

2 Model by Manufacturer

Production Time...ate % by Shift

Data relationships

Raw Material Supplier, Shift, Supervisor Name, Assembly Line, Scrap Units, Units Produced, Defect Rate % and Units Defective

Raw Material Su...	Shift	Supervisor Name	Assembly Line	Scrap Units	Units Produced	Defect Rate %	Units Defective
AutoParts India	Afternoon	Anand P	LINE-1	1	62	3.23	2
		Summary		**1**	**62**	**3.23**	**2**
		Prakash S	LINE-1	4	91	6.59	6
			LINE-2	1	32	9.38	3
			LINE-4	6	135	6.67	9
		Summary		**11**	**258**	**6.98**	**18**
		Rajesh K	LINE-5	1	54	1.85	1
			LINE-6	4	67	7.46	5
			LINE-7	0	49	0	0
		Summary		**5**	**170**	**3.53**	**6**
		Suresh M	LINE-6	1	21	4.76	1
		Summary		**1**	**21**	**4.76**	**1**
		Vijay R	LINE-5	3	62	6.45	4
			LINE-6	4	74	5.41	4
		Summary		**7**	**136**	**5.88**	**8**
	Summary			**25**	**647**	**5.41**	**35**

Details

No details found

No details were found for this visualization.

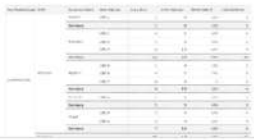
ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=i72AE601D97324CF9A8A954B0B7685710&objRef=i72AE601D97324CF9A8A954B0B768571...

IBM Cognos Analytics | My CS 4

Create | 2/9 | Pie | Related | Compare | Analytics | Details | Fields | Properties

Cards

Raw Material Su...nits Defective



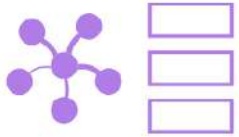
1

Model by Manufacturer

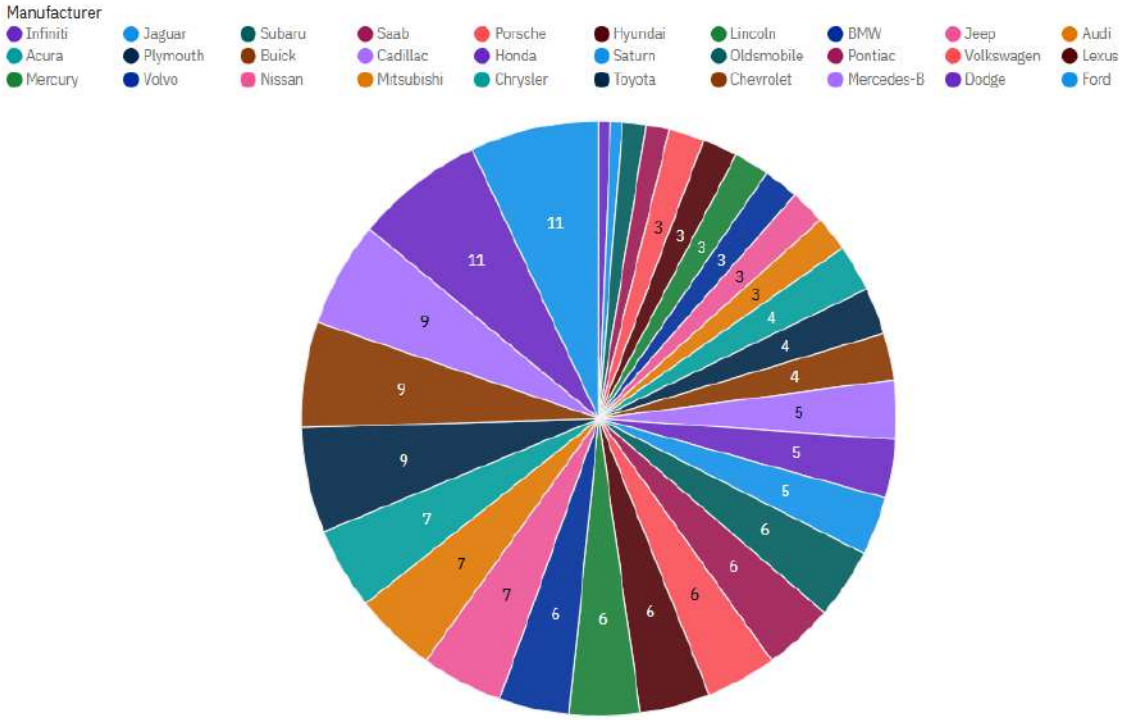
2

Production Time...ate % by Shift

Data relationships



Model by Manufacturer



Details

The total number of results for **Model**, across all **manufacturers**, is 157.

Dodge (7 %) and Ford (7 %) are the most frequently occurring categories of **Manufacturer** with a combined count of 22 items with **Model** values (14 % of the total).

ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=i72AE601D97324CF9A8A954B0B7685710&objRef=i72AE601D97324CF9A8A954B0B768571...

IBM Cognos Analytics | My CS 4

Create | 3/9 | Line and column | Related | Compare | Analytics | Details | Fields | Properties

Cards

1

Model by Manufacturer

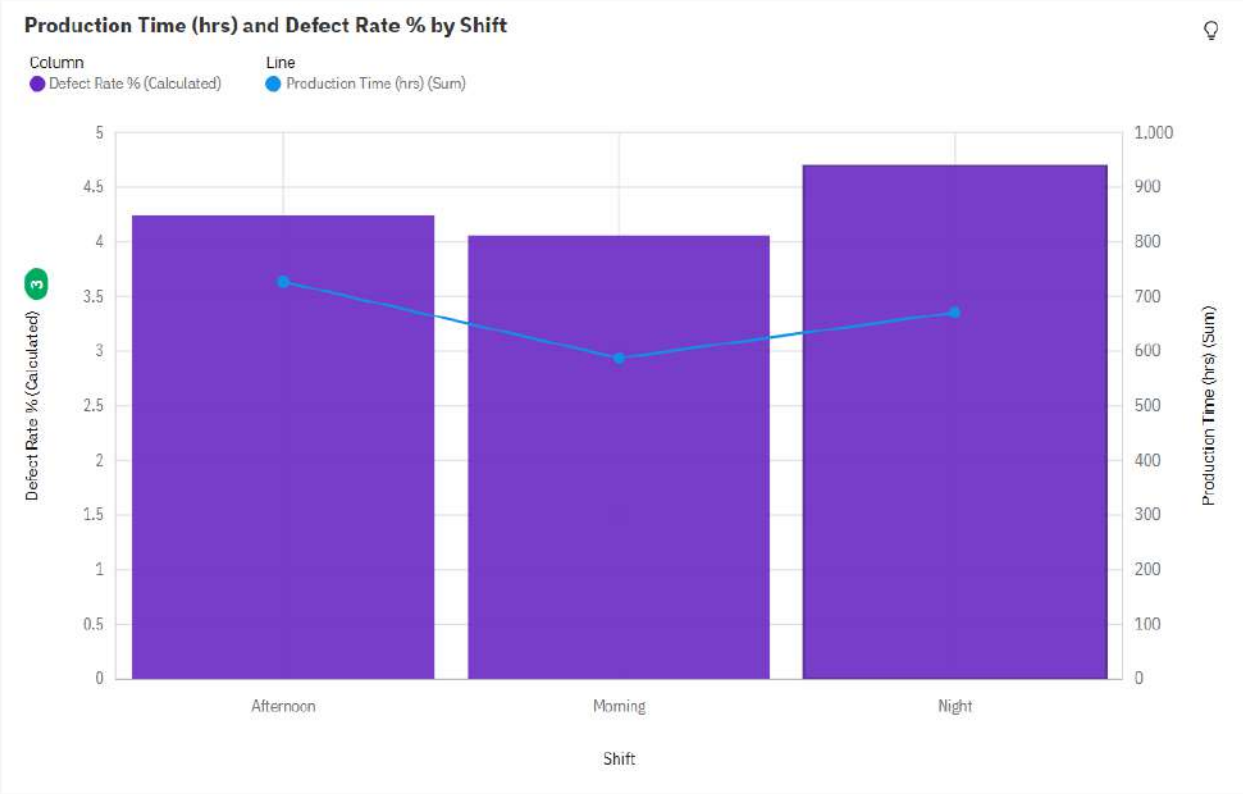
2

Production Time...ate % by Shift

3

Defect Rate % ...pervisor Name

Data relationships



Details

The total number of results for **Defect Rate %**, across all **shifts**, is 200.

Afternoon (36.5 %) and Night (33.5 %) are the most frequently occurring categories of **Shift** with a combined count of 140 items with **Defect Rate %** values (70 % of the total).

Over all **shifts**, the sum of **Production Time (hrs)** is nearly two thousand.

Production Time (hrs) ranges from 587.3, when **Shift** is Morning, to 726.9, when **Shift** is Afternoon.

For **Production Time (hrs)**, the most significant values of **Shift** are Afternoon and Night, whose respective **Production Time (hrs)** values add up to almost 1500, or 70.4 % of the total.

ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=i72AE601D97324CF9A8A954B0B7685710&objRef=i72AE601D97324CF9A8A954B0B768571...

IBM Cognos Analytics | My CS 4

Create | 4/9 | Bar | Related | Compare | Analytics | Details | Fields | Properties

Cards

2

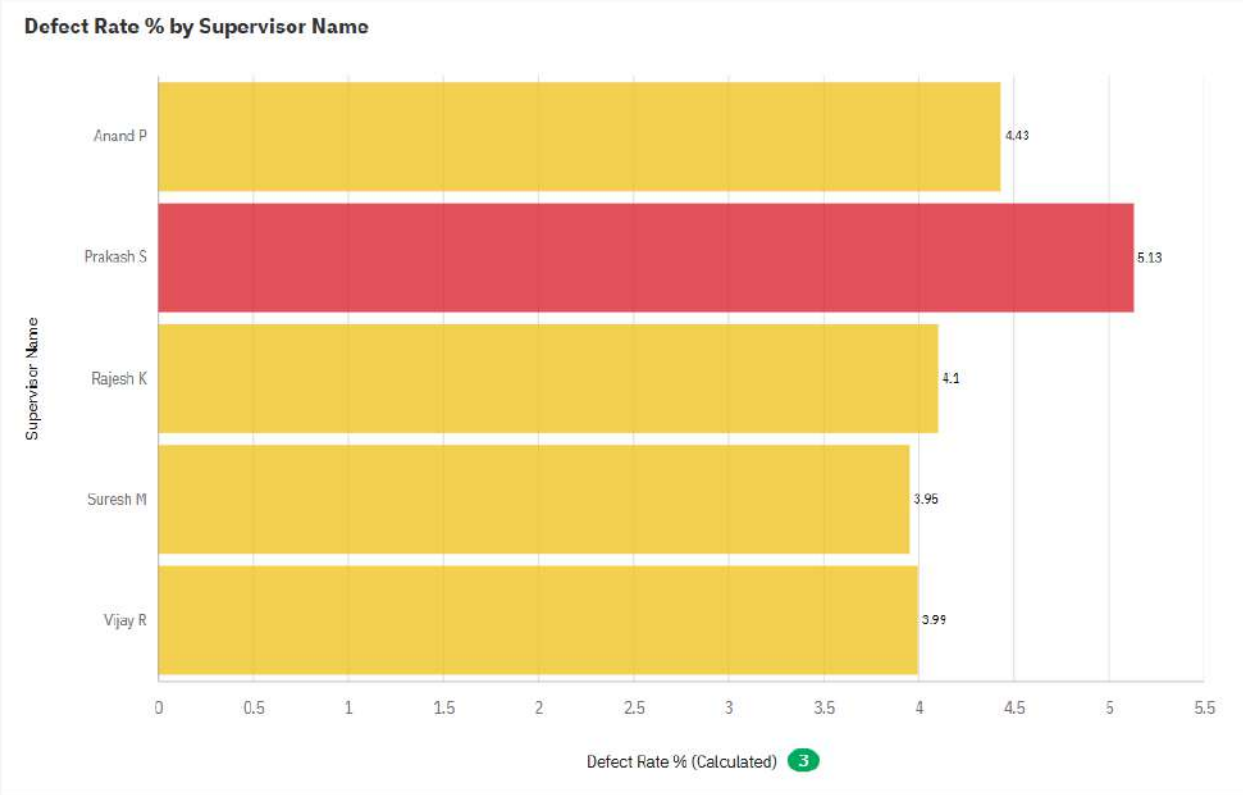
Production Time...ate % by Shift

3

Defect Rate % ...pervisor Name

4

Data relationships



Details

The total number of results for **Defect Rate %**, across all **supervisor names**, is 200.

ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=i72AE601D97324CF9A8A954B0B7685710&objRef=i72AE601D97324CF9A8A954B0B768571...

IBM Cognos Analytics | My CS 4

Create + | 5/9 | Column | Related | Compare | Analytics | Details | Fields | Properties

Cards

Defect Rate % ...pervisor Name

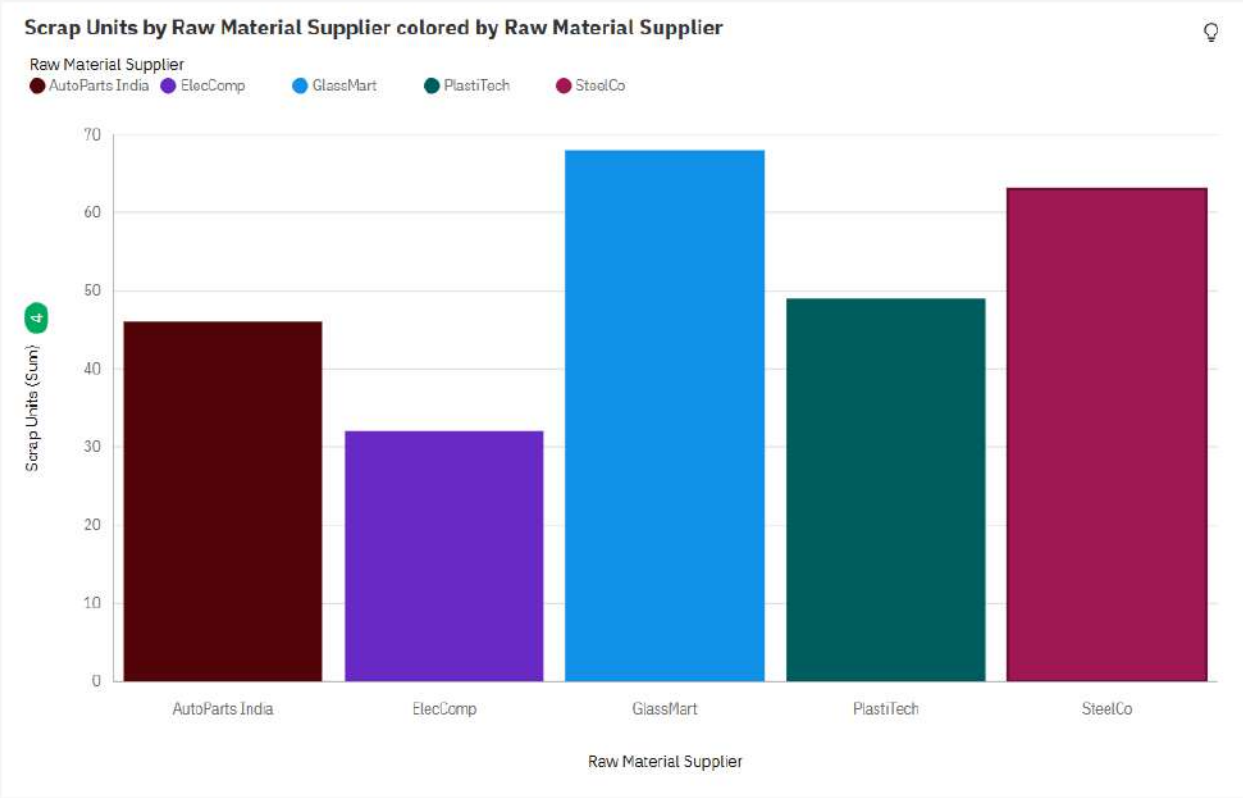
4

Scrap Units by R...terial Supplier

5

Scrap Units an...roduction Date

Data relationships



Details

Over all **raw material suppliers** and **raw material suppliers**, the sum of **Scrap Units** is 258.

The summed values of **Scrap Units** range from 32 to 68.

For **Scrap Units**, the most significant values of **Raw Material Supplier** are GlassMart and SteelCo, whose respective **Scrap Units** values add up to 131, or 50.8 % of the total.

ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=i72AE601D97324CF9A8A954B0B7685710&objRef=i72AE601D97324CF9A8A954B0B768571...

IBM Cognos Analytics | My CS 4

Create + | 6/9 | Line | Related | Compare | Analytics | Details | Fields | Properties

Cards

5

Scrap Units an...roduction Date

6

Defect Rate % ... Assembly Line

7

Supervisor Name

Data relationships

Scrap Units and Material Cost per Unit by Production Date

Measures

Material Cost per Unit | Scrap Units

Values

550,000
500,000
450,000
400,000
350,000
300,000
250,000
200,000
150,000
100,000
50,000
0

1/2/2024 1/9/2024 1/15/2024 1/23/2024 2/1/2024 2/7/2024 2/15/2024 2/24/2024 3/1/2024 3/5/2024 3/17/2024 3/26/2024 4/1/2024 4/8/2024 4/14/2024 4/20/2024 4/30/2024 5/9/2024 5/17/2024 5/23/2024 6/2/2024 6/11/2024 6/18/2024 6/29/2024 7/12/2024 7/23/2024 8/6/2024 8/15/2024 8/26/2024 9/3/2024 9/16/2024 9/25/2024 10/1/2024 10/9/2024 10/17/2024 10/31/2024 11/7/2024 11/22/2024 12/7/2024 12/18/2024 12/30/2024

Production Date

Details

The total number of results for **Scrap Units**, across all **production dates**, is 200.

Over all **production dates**, the average of **Scrap Units** is 1.29.

The total number of results for **Material Cost per Unit**, across all **production dates**, is 200.

Across all **production dates**, the average of **Material Cost per Unit** is over 120 thousand.

Material Cost per Unit ranges from over 50 thousand, when **Production Date** is 2024-06-18, to almost 513 thousand, when **Production Date** is 2024-02-08.

Scrap Units ranges from 0, when **Production Date** is 2024-01-02, to 11, when **Production Date** is 2024-02-03.

2024-07-03 (1.5 %), 2024-06-23 (1.5 %), 2024-03-21 (1.5 %), 2024-05-17 (1.5 %), and 2024-11-06 (1.5 %) are the most frequently occurring categories of

ap2.ca.analytics.ibm.com/bi/?perspective=explore&id=i72AE601D97324CF9A8A954B0B7685710&objRef=i72AE601D97324CF9A8A954B0B768571...

IBM Cognos Analytics | My CS 4

Create + | 7/9 | Column | Related | Compare | Analytics | Details | Fields | Properties

Cards

7

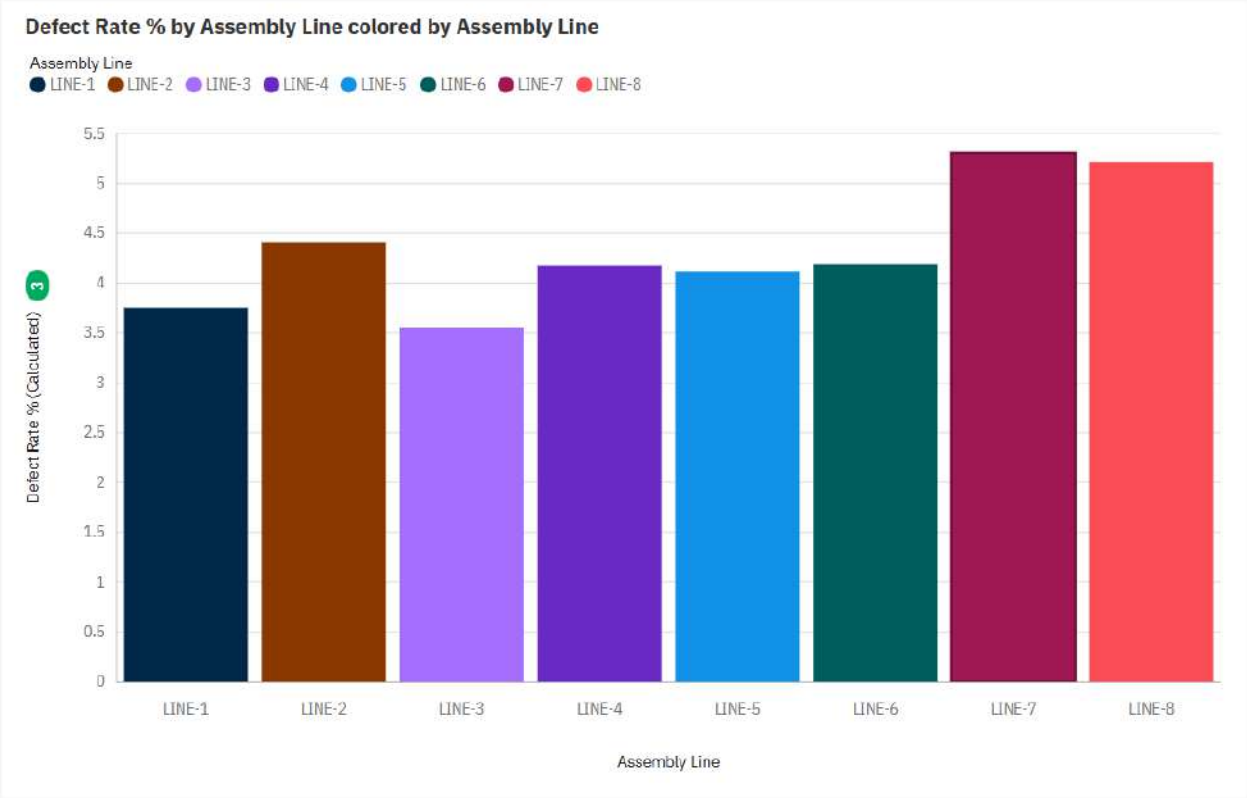
Supervisor Name

8

Supervisor Name

9

Data relationships



Details

The total number of results for **Defect Rate %**, across all **assembly lines**, is 200.

LINE-8 (16 %) and LINE-3 (15 %) are the most frequently occurring categories of **Assembly Line** with a combined count of 62 items with **Defect Rate %** values (31 % of the total).